

AfriShield Hardware Demo

Cost Breakdown — gate + wall + pump action rig, in USD

Covers the physical “detect → decide → act” demo: one ESP32, one water-level sensor, two servos (diversion gate + flood wall), one drainage pump, status LEDs, and wiring/model materials. This is the same design already verified working end-to-end in the Wokwi simulation (hardware/wokwi-action-demo/).

Exchange rate used: 1 USD = 50.95 EGP (2026-09-08, xe.com/Wise mid-market rate) — local Egyptian-market prices converted for reference; actual cost in EGP may vary slightly day to day.

Itemized Cost

Component	Price (USD)	Price (EGP)	Purpose	Source
ESP32 dev board (CH340, USB-C)	\$5.89	300 EGP	Main controller — WiFi-capable, replaces an Arduino Uno	Micro Ohm Electronics
HC-SR04 ultrasonic sensor	\$0.79	40 EGP	Water-level input (mounted above water, pointing down — dry, non-waterproof is fine for a demo)	Micro Ohm Electronics
SG90 micro servo ×2	\$3.34	170 EGP	Diversion gate + flood wall actuators	Most Electronics
Mini submersible DC pump (3–6V)	\$0.88	45 EGP	Drainage action	Micro Ohm Electronics
1-channel 5V relay module	\$1.67	85 EGP	Switches the pump on/off from the ESP32	Amazon.eg
2× LEDs (green/red) + resistors *	\$0.16-0.29	8-15 EGP	Status indicators — normal / high water	typical hobby-store price
4×AA battery holder + batteries *	\$0.59-0.98	30-50 EGP	Separate power for the 2 servos + pump, to avoid browning out the ESP32	Micro Ohm Electronics (holder)
Jumper wires + small breadboard *	\$0.98-1.57	50-80 EGP	Wiring	typical hobby-store price
Model materials (cardboard/tubing/glue) *	\$0.79-1.18	40-60 EGP	The physical water rig: gate channel, wall, pump basin	craft/hobby materials

* Not individually price-checked against a specific store — estimated from typical hobby-electronics/craft pricing. Everything else is a real, sourced price from an Egyptian retailer.

Verified subtotal (sourced prices)	\$12.56	640 EGP
Estimated items (unsourced)	\$2.51-4.02	128-205 EGP
Subtotal	\$15.07-16.58	768-845 EGP
+15% buffer (price/currency fluctuation)	\$2.26-2.49	115-127 EGP
Realistic total, with buffer	\$17.33-19.07	883-972 EGP

Service Cost (non-hardware)

Africa's Talking's free sandbox never delivers to a real phone under any circumstances (SMS, Voice, and USSD all route to their internal simulator only) — confirmed directly from Africa's Talking's own help center. A real SMS/voice demo needs a **Live** app with real prepaid credit instead. Real SMS costs ~\$0.01–0.03/message, so this covers roughly 100+ real demo messages, not a subscription fee.

Item	Cost (USD)	Purpose
Africa's Talking Live app credit	\$25.00	Prepaid balance so real SMS/voice can actually reach a phone during the demo — sandbox alone cannot do this

Grand total (hardware + Africa's Talking credit)	\$42.33-44.07
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Design notes:

- Rain gauge intentionally excluded from this rig — physical actuation (gate/wall/pump) reacts to the confirmed, current water level, not a predictive rainfall signal. Rainfall still feeds the separate AI risk-scoring pipeline (hardware/wokwi-flood-sensor/).
- Power the ESP32 itself via a USB cable from a laptop or power bank — no added cost, you almost certainly already have one. Give the 2 servos + pump their own separate battery pack instead of sharing that same 5V line, which risks the board resetting when a motor draws a current spike.
- Plain HC-SR04 (non-waterproof) is fine here since the sensor stays dry, mounted above the water pointing down — waterproofing (JSN-SR04T) only matters for a real outdoor field deployment, not this tabletop demo.